

Generalized Autoregressive Conditional Heteroskedasticity

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ANALYTICAL ABSTRACT

Engle (1982) made the conditional variance of a time series a function of past squared shocks, but practical ARCH models needed long lag lengths whose free estimation violated non-negativity, so applied work imposed ad hoc declining lag weights. Bollerslev adds lagged conditional variances to the variance equation, producing the GARCH(p,q) process. The extension mirrors the step from AR to ARMA: a geometric lag in past squared shocks is obtained from three parameters. The paper derives the necessary and sufficient condition for covariance stationarity, $A(1) + B(1) < 1$, the exact moment conditions and kurtosis of the GARCH(1,1) process, an ARMA representation for the squared process that justifies identification through its autocorrelations, maximum likelihood estimation with a block-diagonal information matrix, and the result that Lagrange multiplier tests against a full GARCH alternative are infeasible under a white-noise null. Applied to quarterly U.S. GNP deflator inflation from 1948 to 1983, a GARCH(1,1) model fits marginally better than the ARCH(8) specification of Engle and Kraft (1983), passes residual diagnostics, and implies a mean lag of about six quarters in the transmission of inflation uncertainty.

1 Introduction

1.1 Paper Identification

Table 1. Bibliographic and classification record

Field	Entry
Exact title	Generalized Autoregressive Conditional Heteroskedasticity
Author	Tim Bollerslev (University of California at San Diego; Institute of Economics, University of Aarhus)
Publication	Journal of Econometrics 31 (1986), 307–327. Received May 1985, final version February 1986
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Field / subfield	Econometrics / time-series models of conditional heteroskedasticity
Series	Time Series
Paper type	Methodological and econometric (probability theory of a new process, estimation, testing), with a short empirical application

Source: the paper itself; classification by the reviewer.

1.2 Executive Understanding

WHAT THE PAPER IS TRYING TO DISCOVER

The paper asks whether the conditional variance of a time series can be modelled with a lag structure that is long in memory but short in parameters, and what the statistical properties of such a model are: when it is stationary, which moments exist, how it can be identified from data, how it can be estimated, and how its presence can be tested.

WHY THE QUESTION MATTERS

Forecast uncertainty is itself an economic object. Risk premia, option values, optimal hedge ratios, the welfare cost of inflation and the width of a central bank's forecast fan all depend on a conditional second moment. A model that treats the variance as constant gives correct point forecasts but systematically wrong interval forecasts whenever volatility clusters, which it does in almost every economic and financial series.

WHAT WAS KNOWN BEFORE

Engle (1982) had introduced the ARCH(q) process, in which the conditional variance is a linear function of q past squared innovations, and had shown its usefulness for U.K. inflation. Weiss (1982, 1984) developed asymptotic theory and ARMA models with ARCH errors, and Milhøj (1984) studied ARCH moment structure. In every empirical application, however, the long lags required by the data were handled by imposing a fixed, linearly declining weight pattern, because a freely estimated long lag routinely produced negative coefficients.

WHAT WAS MISSING

A parsimonious parametric family that could deliver long memory in the variance without arbitrary restrictions, together with the probability theory needed to use it: stationarity, moments, an identification device and a likelihood theory.

WHAT THE PAPER PROPOSES AND HOW

The paper adds p lags of the conditional variance itself to the ARCH equation. It then proves a covariance-stationarity theorem, derives exact moment conditions for the GARCH(1,1) case, establishes an ARMA representation for the squared process, adapts Engle's maximum likelihood theory and LM tests, and illustrates the model on U.S. inflation.

WHAT IT FINDS AND WHY IT MATTERS

A three-parameter GARCH(1,1) variance equation matches or beats a heavily restricted ARCH(8) on U.S. inflation and passes the same diagnostics. The deeper finding is structural: the conditional variance can be written as an ARCH(∞) process with geometrically declining weights, so the GARCH recursion is the variance analogue of the ARMA model. This single idea made conditional volatility modelling routine in finance, and GARCH(1,1) remains, four decades later, the benchmark that more elaborate volatility models must beat.

2 Research Problem and Hypothesis

2.1 Research Question

The precise question is: can the ARCH conditional variance equation be generalised by including lagged conditional variances so that (i) the implied lag distribution on past squared innovations is flexible, long and non-negative with few parameters; (ii) the resulting process has a tractable stationarity and moment structure; (iii) it can be identified, estimated and tested with standard tools; and (iv) on the U.S. inflation series previously modelled by Engle and Kraft (1983), it yields a fit at least as good as the restricted ARCH(8) with a more plausible lag structure?

2.2 Motivation

The motivation is practical before it is theoretical. Engle (1982), Engle (1983) and Engle and Kraft (1983) all needed long ARCH lags and all imposed a linear declining lag structure because unrestricted estimation violated the non-negativity constraints. Bollerslev states the analogy explicitly: as the ARMA model offers a parsimonious representation of a long autoregression, so the GARCH model should offer a parsimonious representation of a long ARCH. The economic motivation is the interpretation of the lagged variance term as an adaptive learning mechanism: agents update their assessment of uncertainty partly from yesterday's assessment and partly from yesterday's surprise.

2.3 Existing Literature

The relevant prior work falls into three strands. The first is the ARCH model and its applications: Engle (1982) for U.K. inflation; Engle (1983) and Engle and Kraft (1983) for U.S. inflation; Coulson and Robins (1985) relating inflation volatility to activity; Engle, Lilien and Robins (1985) using a conditional variance as a term-premium proxy; Domowitz and Hakkio (1985) for foreign exchange; and Weiss (1984), who found ARMA models with ARCH errors successful for thirteen U.S. macroeconomic series. The second is the statistical theory of ARCH: Weiss (1982) on asymptotics and Milhøj (1984) on moments. The third is classical ARMA methodology: Box and Jenkins (1976) for identification through autocorrelations; Granger and Andersen (1978) for the observation that squared residuals can be autocorrelated when residuals are not; McLeod and Li (1983) for squared-residual diagnostics; and Godfrey (1978) for the breakdown of LM tests against full ARMA alternatives.

2.4 Research Gap

No model in 1985 combined a long, smoothly declining, non-negative lag on past squared shocks with a small number of freely estimated parameters. Equally absent were stationarity and moment results for such a model, and any statement about whether its presence could be tested.

2.5 Contribution and Hypothesis

PAPER The paper claims four contributions: the GARCH(p,q) definition; the stationarity theorem; the autocorrelation-based identification tools; and the extension of ML estimation and LM testing, with a demonstration that GARCH(1,1) beats the ARCH(8) benchmark. The implicit empirical hypothesis is that the long memory of inflation uncertainty is better captured by a geometric lag generated by a lagged conditional variance than by a truncated linear lag.

CRITIQUE The theoretical contributions are fully demonstrated, with proofs. The empirical claim of superiority is modest and honestly so: the paper says GARCH provides a “marginally better fit” and rests its case mostly on lag plausibility and diagnostics, not on a formal non-nested comparison. The lasting contribution is the parametric family itself, whose value was established by the subsequent literature rather than by the single inflation example.

3 Theoretical Framework and Mechanism

3.1 The GARCH(p,q) Process

Let ε_t be a real-valued discrete-time stochastic process and ψ_t the information set (σ -field) of all information through time t . The GARCH(p,q) process is defined by

$$\varepsilon_t | \psi_{t-1} \sim N(0, h_t), \quad (1)$$

$$h_t = \alpha_0 + \sum_{i=1}^q \alpha_i \varepsilon_{t-i}^2 + \sum_{i=1}^p \beta_i h_{t-i} = \alpha_0 + A(L) \varepsilon_t^2 + B(L) h_t, \quad (2)$$

with $p \geq 0, q > 0, \alpha_0 > 0, \alpha_i \geq 0$ for $i = 1, \dots, q$ and $\beta_i \geq 0$ for $i = 1, \dots, p$. In the regression version the ε_t are innovations of a linear model,

$$\varepsilon_t = y_t - x_t' b. \quad (3)$$

h_t conditional variance of ε_t given ψ_{t-1} ; known one period in advance, hence a one-step forecast variance.

α_0 intercept; positivity guarantees a strictly positive floor for the variance.

α_i ARCH coefficients: the reaction of today's variance to the squared surprise of i periods ago.

β_i GARCH coefficients: the weight on the variance that was itself forecast i periods ago.

$A(L), B(L)$ lag polynomials $\sum \alpha_i L^i$ and $\sum \beta_i L^i$; $A(1)$ and $B(1)$ are their sums.

PAPER For $p = 0$ the process reduces to ARCH(q); for $p = q = 0$, ε_t is white noise. The ARCH(q) variance is a function of past sample variances only, while GARCH allows lagged conditional variances to enter, which the author describes as an adaptive learning mechanism. Footnote 1 notes that normality is assumed only to follow Engle (1982) and that other conditional distributions, or absolute rather than squared innovations (McCulloch, 1983), could be used.

INFERENCE The economic content of (2) is easiest to see in the GARCH(1,1) case written as $h_t = h_{t-1} + \alpha_1(\varepsilon_{t-1}^2 - h_{t-1}) + [\alpha_0 - (1 - \alpha_1 - \beta_1)h_{t-1}]$. The second term is an error-correction on the variance forecast: the forecast is revised upward when the realised squared shock exceeds what was expected, with gain α_1 . The third term pulls the forecast toward its long-run level at speed $1 - \alpha_1 - \beta_1$. This is precisely the structure of adaptive expectations applied to a second moment, and it explains why the author speaks of learning.

3.2 The ARCH(∞) Representation

If all roots of $1 - B(z) = 0$ lie outside the unit circle, (2) can be inverted into a distributed lag of past squared innovations:

$$h_t = \alpha_0 (1 - B(1))^{-1} + A(L)(1 - B(L))^{-1} \varepsilon_t^2 = \alpha_0 (1 - \sum_{i=1}^p \beta_i)^{-1} + \sum_{i=1}^{\infty} \delta_i \varepsilon_{t-i}^2, \quad (4)$$

where the δ_i are the coefficients of the power series $D(L) = A(L)(1 - B(L))^{-1}$, obtained recursively as

$$\delta_i = \alpha_i + \sum_{j=1}^n \beta_j \delta_{i-j} \quad (i = 1, \dots, q), \quad \delta_i = \sum_{j=1}^n \beta_j \delta_{i-j} \quad (i = q+1, \dots), \quad (5)$$

with $n = \min\{p, i - 1\}$. The derivation is the usual polynomial identity: multiplying $D(L)(1 - B(L)) = A(L)$ out and matching coefficients on L^i gives $\delta_i - \sum \beta_j \delta_{i-j} = \alpha_i$, with $\alpha_i = 0$ beyond q . For GARCH(1,1) the recursion collapses to $\delta_i = \alpha_1 \beta_1^{i-1}$, a geometric lag whose total weight is $D(1) = \alpha_1 / (1 - \beta_1)$.

PAPER If $B(1) < 1$ the δ_i decrease for $i > \max\{p, q\}$, so when $D(1) < 1$ a GARCH(p,q) can be approximated to any accuracy by a stationary ARCH(Q) with Q large enough. The case for GARCH is therefore parsimony, as with ARMA versus long AR models, not the generation of dynamics an ARCH could never produce.

INFERENCE Equation (4) is the core mechanism of the paper. It shows that the problem with long ARCH models was never the length of memory but the number of free coefficients; GARCH delivers infinite memory with $p + q + 1$ parameters, and non-negativity of every δ_i is automatic under the sufficient constraints $\alpha_i, \beta_i \geq 0$.

3.3 Theorem 1: Covariance Stationarity

THEOREM 1 (BOLLERSLEV, 1986)

The GARCH(p,q) process defined by (1) and (2) is wide-sense stationary with $E(\varepsilon_t) = 0$, $\text{var}(\varepsilon_t) = \alpha_0(1 - A(1) - B(1))^{-1}$ and $\text{cov}(\varepsilon_t, \varepsilon_s) = 0$ for $t \neq s$ if and only if $A(1) + B(1) < 1$.

The proof (appendix A.1) follows Milhøj (1984). Writing $\varepsilon_t = \eta_t h_t^{1/2}$ with η_t i.i.d. $N(0,1)$ and substituting the variance equation into itself repeatedly gives

$$h_t = \alpha_0 \sum_{k=0}^{\infty} M(t, k), \quad M(t, 0) = 1, \quad (A.2)$$

$$M(t, k+1) = \sum_{i=1}^q \alpha_i \eta_{t-i}^2 M(t-i, k) + \sum_{i=1}^p \beta_i M(t-i, k), \quad (A.3)$$

where $M(t, k)$ collects all products of k coefficients and the associated squared standardised shocks. Because η^2 is i.i.d. with unit mean and independent of $M(t-i, k)$, taking expectations gives $E M(t, k+1) = (A(1) + B(1)) E M(t, k)$, hence $E M(t, k) = (A(1) + B(1))^k$ and

$$E(\varepsilon_t^2) = \alpha_0 \sum_{k=0}^{\infty} (A(1) + B(1))^k = \alpha_0 (1 - A(1) - B(1))^{-1}, \quad (A.5)$$

which is finite if and only if $A(1) + B(1) < 1$. The zero mean and zero autocovariance follow from the symmetry of the conditional normal distribution.

INFERENCE The theorem identifies persistence, $A(1) + B(1)$, as the single number that governs the long-run behaviour of the variance. The unconditional variance $\alpha_0 / (1 - A(1) - B(1))$ explodes as persistence approaches one, which is why near-unit persistence estimates in financial data later prompted the integrated GARCH of Engle and Bollerslev (1986). The theorem concerns second-order stationarity only: Nelson (1990) later showed that a GARCH(1,1) can be strictly stationary even when $\alpha_1 + \beta_1 \geq 1$, provided $E \ln(\beta_1 + \alpha_1 \eta_t^2) < 0$.

3.4 The ARMA Representation of the Squared Process

Following a suggestion credited to Sastry Pantula and a referee, define $v_t = \varepsilon_t^2 - h_t = (\eta_t^2 - 1)h_t$. Adding $\varepsilon_t^2 - h_t$ to both sides of (2) gives

$$\varepsilon_t^2 = \alpha_0 + \sum_{i=1}^q \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^p \beta_j \varepsilon_{t-j}^2 - \sum_{j=1}^p \beta_j v_{t-j} + v_t \quad (6)$$

$$v_t = (\eta_t^2 - 1) h_t, \quad \eta_t \sim i.i.d. N(0, 1). \quad (7)$$

Since $E(v_t | \psi_{t-1}) = 0$, v_t is a martingale difference and hence serially uncorrelated. Equation (6) is therefore an ARMA(m, p) model in ε_t^2 with $m = \max\{p, q\}$, autoregressive coefficients $\alpha_i + \beta_i$ and moving-average coefficients $-\beta_j$.

CRITIQUE The representation is exact but its innovation v_t is conditionally heteroskedastic, since its conditional variance is $2h_t^2$ under normality. Linear ARMA estimation on squared data is therefore consistent only under fourth-moment conditions and highly inefficient, which is why the paper rightly uses (6) for identification rather than estimation.

3.5 The GARCH(1,1) Process and Theorem 2

The workhorse specification is

$$h_t = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 h_{t-1}, \quad \alpha_0 > 0, \alpha_1 \geq 0, \beta_1 \geq 0. \quad (8)$$

The printed equation (8) omits the plus sign between α_0 and $\alpha_1 \varepsilon_{t-1}^2$; the version above is the one implied by (2) and used throughout the paper.

THEOREM 2 (BOLLERSLEV, 1986)

For the GARCH(1,1) process a necessary and sufficient condition for existence of the $2m$ -th moment is $\mu(\alpha_1, \beta_1, m) < 1$, and the $2m$ -th moment then satisfies the recursion (11).

$$\mu(\alpha_p, \beta_p, m) = \sum_{j=0}^m C(m, j) a_j \alpha_1^j \beta_1^{m-j} < 1, \quad a_0 = 1, \quad a_j = \prod_{i=1}^j (2i - 1), \quad (9-10)$$

$$E(\varepsilon_t^{2m}) = a_m \left[\sum_{n=0}^{m-1} a_n^{-1} E(\varepsilon_t^{2n}) \alpha_0^{m-n} C(m, m-n) \mu(\alpha_p, \beta_p, n) \right] [1 - \mu(\alpha_p, \beta_p, m)]^{-1}. \quad (11)$$

Here $a_j = E(\eta^{2j})$ is the $2j$ -th moment of a standard normal (1, 3, 15, 105, ...) and $C(m, j)$ is the binomial coefficient. The logic of the proof (appendix A.2) is transparent. By normality $E(\varepsilon_t^{2m}) = a_m E(h_t^m)$. Expanding $h_t^m = (\alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 h_{t-1})^m$ binomially and using $E(\varepsilon_{t-1}^{2j} | \psi_{t-2}) = a_j h_{t-1}^j$ gives

$$E(h_t^m | \psi_{t-2}) = \sum_{n=0}^m h_{t-1}^n C(m, n) \alpha_0^{m-n} \sum_{j=0}^n C(n, j) a_j \alpha_1^j \beta_1^{n-j}. \quad (A.8)$$

Stacking $w_t = (h_t^m, h_t^{m-1}, \dots, h_t)'$ gives a linear system $E(w_t | \psi_{t-2}) = d + C w_{t-1}$ with C upper triangular and diagonal elements $\mu(\alpha_1, \beta_1, i)$. Iterating backwards, the limit $(I - C)^{-1}d$ exists if and only if all eigenvalues, which are the diagonal elements, lie inside the unit circle. The paper notes that $\mu(\alpha_1,$

$\beta_1, i) < 1$ implies the same inequality at $i - 1$, so the binding condition is the one for the highest moment. This also follows from writing $\mu(\alpha_1, \beta_1, m) = E[(\beta_1 + \alpha_1 \eta_1^2)^m]$ and applying Lyapunov's inequality.

For $m = 1$, $\mu = \alpha_1 + \beta_1$, recovering Theorem 1. For $m = 2$, $\mu = \beta_1^2 + 2\alpha_1\beta_1 + 3\alpha_1^2$, which yields

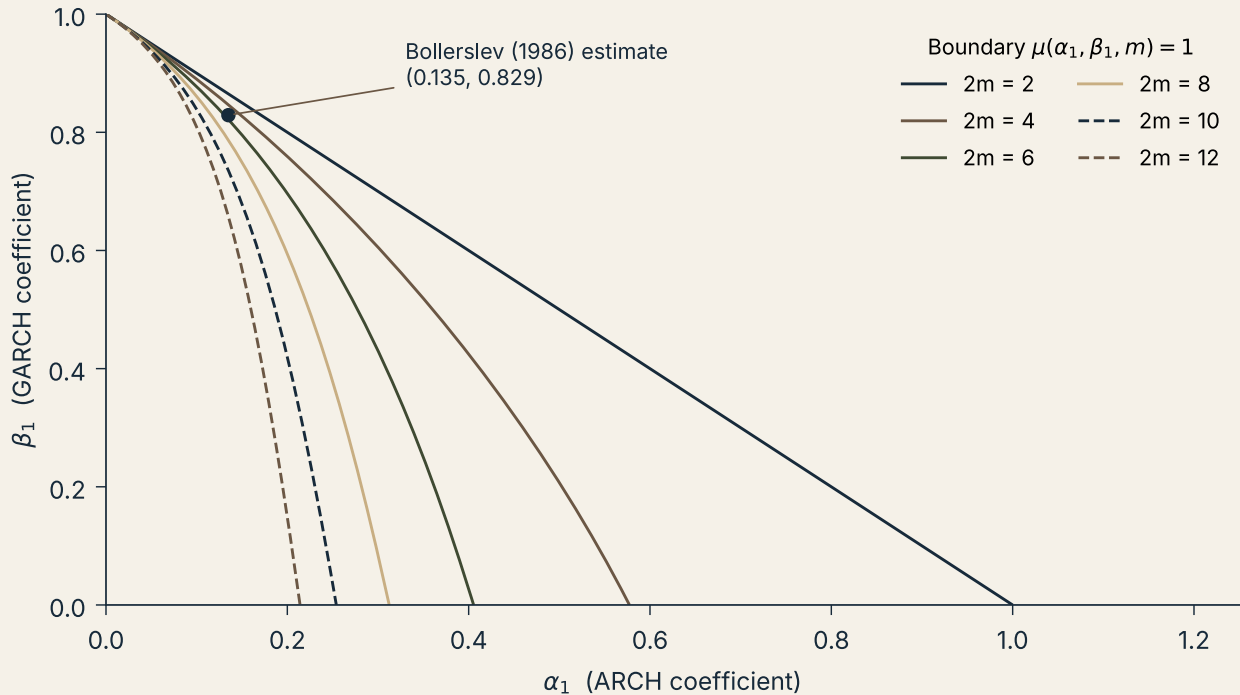
$$E(\varepsilon_1^4) = 3\alpha_1^2(1 + \alpha_1 + \beta_1) / [(1 - \alpha_1 - \beta_1)(1 - \beta_1^2 - 2\alpha_1\beta_1 - 3\alpha_1^2)], \quad (\text{K1})$$

$$K = [E(\varepsilon_1^4) - 3E(\varepsilon_1^2)^2] E(\varepsilon_1^2)^{-2} = 6\alpha_1^2(1 - \beta_1^2 - 2\alpha_1\beta_1 - 3\alpha_1^2)^{-1}. \quad (\text{K2})$$

PAPER Excess kurtosis is strictly positive whenever $\alpha_1 > 0$: the GARCH(1,1) process is leptokurtic even though its conditional distribution is normal. For $\beta_1 = 0$, (9) reduces to the ARCH(1) condition $\alpha_1 < 1$; and because of its longer memory, a GARCH(1,1) whose total lag weight $\alpha_1/(1 - \beta_1)$ exceeds the ARCH(1) moment bound may still possess the moment.

INFERENCE Equation (K2) is the most consequential formula in the paper for finance. It says that fat tails in unconditional returns can arise purely from time variation in a Gaussian variance, and that what drives the tail is α_1 , the reactivity to news, not persistence as such. Holding $\alpha_1 + \beta_1$ fixed, kurtosis rises steeply as weight is shifted from β_1 to α_1 , which Figure 4 below illustrates.

Figure 1. Moment-existence regions of the GARCH(1,1) process, boundaries $\mu(\alpha_1, \beta_1, m) = 1$ for $2m = 2$ to 12



Generated by the reviewer from equations (9)–(10) of the paper; this recomputes the content of the paper's Fig. 1. The $2m$ -th moment exists below and to the left of each curve. The marked point is the GARCH(1,1) estimate from equation (31).

Figure 1 makes the trade-off visible. Each higher moment requires a progressively smaller region near the origin, and the curves bend more sharply in the α_1 direction than in the β_1 direction, because α_1 enters multiplied by the normal moments a_j , which grow factorially. The paper's own

inflation estimate sits just inside the fourth-moment boundary ($\mu = 0.966$) and outside the sixth-moment boundary ($\mu = 1.021$, reviewer's computation), so the fitted process has a finite kurtosis but an infinite sixth moment.

3.6 Lag Structure: Mean and Median Lag

With $\delta_i = \alpha_1 \beta_1^{i-1}$, the mean and median lags of the conditional variance equation (Harvey, 1982) are

$$\zeta = \sum_{i=1}^{\infty} i \delta_i / \sum_{i=1}^{\infty} \delta_i = (1 - \beta_1)^{-1}, \quad \rho = -\log 2 / \log \beta_1. \quad (L)$$

Both depend only on β_1 : the GARCH coefficient sets the shape of the memory, the ARCH coefficient sets its scale. This separation of roles is the mechanical reason why GARCH(1,1) is so flexible with only two dynamic parameters.

3.7 Autocorrelation and Partial Autocorrelation of ε_t^2

Assuming finite fourth moments, let $\gamma_n = \text{cov}(\varepsilon_t^2, \varepsilon_{t-n}^2)$. From (6) and (7),

$$\gamma_n = \sum_{i=1}^q \alpha_i \gamma_{n-i} + \sum_{i=1}^p \beta_i \gamma_{n-i} = \sum_{i=1}^m \varphi_i \gamma_{n-i}, \quad n \geq p+1, \quad \varphi_i = \alpha_i + \beta_i, \quad (13)$$

$$\rho_n = \gamma_n / \gamma_0 = \sum_{i=1}^m \varphi_i \rho_{n-i}, \quad n \geq p+1. \quad (14)$$

The first p autocorrelations depend directly on all parameters, after which the Yule-Walker-type recursion (14) determines all higher-lag autocorrelations, exactly as in an ARMA(m , p) process. The partial autocorrelations φ_{kk} , defined by solving $\rho_n = \sum_{i=1}^k \varphi_{ki} \rho_{n-i}$ for $n = 1, \dots, k$ (equation 15), cut off after lag q for a pure ARCH(q) (equation 16) and decay without cutting off for a GARCH(p, q). Sample analogues, with asymptotic variance $1/T$ under the null of no GARCH (Weiss, 1984; McLeod and Li, 1983), serve as identification and diagnostic tools.

INFERENCE This section gives applied researchers the Box-Jenkins toolkit for the second moment. A slowly decaying ACF of squared residuals with a non-truncating PACF is the signature of GARCH; a PACF that truncates at q points to ARCH(q). The practical value is large, but the tools inherit two weaknesses: they require finite fourth moments, which Figure 1 shows is restrictive, and sample autocorrelations of squares converge slowly or not at all when fourth moments are near the boundary.

3.8 Maximum Likelihood Estimation

Let $z'_t = (1, \varepsilon_{t-1}^2, \dots, \varepsilon_{t-q}^2, h_{t-1}, \dots, h_{t-p})$, $\omega' = (\alpha_0, \alpha_1, \dots, \alpha_q, \beta_1, \dots, \beta_p)$ and $\theta = (b', \omega')$ in a compact parameter set with the truth in its interior. The model is $\varepsilon_t = y_t - x'_t b$, $\varepsilon_t | \psi_{t-1} \sim N(0, h_t)$, $h_t = z'_t \omega$ (equation 17), and the average log likelihood, conditional on pre-sample values, is

$$L_T(\theta) = T^{-1} \sum_{t=1}^T l_t(\theta), \quad l_t(\theta) = -\frac{1}{2} \log h_t - \frac{1}{2} \varepsilon_t^2 h_t^{-1}. \quad (18)$$

The score with respect to the variance parameters is

$$\partial l / \partial \omega = \frac{1}{2} h_t^{-1} (\partial h_t / \partial \omega) (\varepsilon_t^2 h_t^{-1} - 1), \quad \partial h_t / \partial \omega = z_t + \sum_{i=1}^p \beta_i \partial h_{t-i} / \partial \omega, \quad (19, 21)$$

and with respect to the mean parameters

$$\partial l / \partial b = \varepsilon_t x_t h_t^{-1} + \frac{1}{2} h_t^{-1} (\partial h_t / \partial b) (\varepsilon_t^2 h_t^{-1} - 1), \quad (22)$$

$$\partial h_t / \partial b = -2 \sum_{j=1}^q \alpha_j x_{t-j} \varepsilon_{t-j} + \sum_{j=1}^p \beta_j \partial h_{t-j} / \partial b. \quad (24)$$

The only difference from Engle (1982) is the recursive term in (21) and (24), which is stable when $B(1) < 1$. The variance score has the structure of a weighted regression of the standardised squared residual minus one on the variance gradient; its conditional mean is zero because $E(\varepsilon_t^2 h_t^{-1} | \psi_{t-1}) = 1$. The off-diagonal block of the information matrix between b and ω is zero, so the mean and variance parameters can be estimated separately without loss of asymptotic efficiency. The recursion is started at the sample mean of ε_t^2 (footnote 5).

Because the recursive derivatives complicate Engle's auxiliary-regression scoring, the paper uses the Berndt, Hall, Hall and Hausman (1974) algorithm:

$$\theta^{(i+1)} = \theta^{(i)} + \lambda_i (\sum_{t=1}^T (\partial l / \partial \theta) (\partial l / \partial \theta')^{-1} \sum_{t=1}^T \partial l / \partial \theta, \quad (\text{BHHH})$$

where λ_i is a line-search step. The direction is the coefficient vector from regressing a $T \times 1$ vector of ones on the scores. Using Weiss (1982), the MLE is strongly consistent and asymptotically normal with covariance $\mathbf{I}^{-1}$, consistently estimated by the inverse outer product of the scores at the last iteration. Replacing normality by the weaker moment conditions

$$E(\varepsilon_t | \psi_{t-1}) = 0, \quad E(\varepsilon_t^2 h_t^{-1} | \psi_{t-1}) = 1, \quad E(\varepsilon_t^4 h_t^{-2} | \psi_{t-1}) \leq M < \infty, \quad (25)$$

the estimator remains consistent and asymptotically normal but with the sandwich covariance $\mathbf{I}^{-1} \mathbf{F} \mathbf{I}^{-1}$, where $\mathbf{F}$ is the outer-product matrix (Weiss, 1982; White, 1982).

CRITIQUE The asymptotic results are imported from Weiss (1982) rather than proved for GARCH, and Weiss's conditions were developed for ARCH. A rigorous QMLE theory for GARCH(1,1) arrived only with Lee and Hansen (1994) and Lumsdaine (1996), and for general GARCH(p,q) with Berkes, Horváth and Kokoszka (2003) and Francq and Zakoian (2004). The paper's claim is right, but in 1986 it was a conjecture by analogy. Equation (25) is also prescient: it is the quasi-likelihood idea that Bollerslev and Wooldridge (1992) later made standard practice.

3.9 Testing for GARCH

Partitioning the variance equation as $h_t = z'_{1t} \omega_1 + z'_{2t} \omega_2$ (equation 26), the LM statistic for $H_0: \omega_2 = 0$ is

$$\xi_{LM} = \frac{1}{2} f'_0 Z_0 (Z'_0 Z_0)^{-1} Z'_0 f_0 \quad (27)$$

$$f_0 = (\varepsilon_1^2 h_1^{-1} - 1, \dots, \varepsilon_T^2 h_T^{-1} - 1)', \quad Z_0 = (h_1^{-1} \partial h_1 / \partial \omega, \dots, h_T^{-1} \partial h_T / \partial \omega)', \quad (28)$$

evaluated under H_0 and asymptotically $\chi^2(r)$ with $r = \dim(\omega_2)$. It differs from Breusch and Pagan (1978) because $\partial h_t / \partial \omega$ carries the recursive term. Under normality an asymptotically equivalent form is $T \cdot R^2$ from regressing f_0 on Z_0 .

PAPER The central negative result: under a white-noise null, $Z_0' Z_0$ is singular when $p > 0$ and $q > 0$, so a general test for GARCH(p, q) is infeasible. Under an ARCH(q) null it is singular for GARCH($r_1, q + r_2$) alternatives with $r_1, r_2 > 0$, and the LM tests against GARCH(r, q) and ARCH($q + r$) coincide. The parallel is Godfrey (1978) for AR versus MA errors, and the problem afflicts Wald and LR tests as well.

INFERENCE The reason is an identification failure at the null. When $\alpha_1 = 0$, the parameter β_1 has no effect on the likelihood, because $h_t = \alpha_0 + \beta_1 h_{t-1}$ converges to a constant whatever β_1 is; a nuisance parameter that is identified only under the alternative breaks standard asymptotics. This is the Davies (1987) problem, later formalised for GARCH by Andrews (2001). Its practical implication, still valid today, is that one tests for ARCH effects and then estimates GARCH, rather than testing for GARCH directly.

4 Data and Empirical Strategy

4.1 Data

The variable is quarterly U.S. inflation, $\pi_t = 100 \cdot \ln(GD_t / GD_{t-1})$, where GD_t is the implicit price deflator for GNP. Values come from the Citibank Economic Database and the U.S. Department of Commerce Survey of Current Business (Vol. 64, no. 9, September 1984). The estimation sample runs from 1948:2 to 1983:4, 143 observations. Engle and Kraft (1983) used 1948:2 to 1980:3, which accounts for small differences between their estimates and the paper's re-estimation.

Table 2. Data characteristics of the empirical example

Item	Description
Series	Implicit GNP price deflator, United States
Transformation	Quarterly log-difference, multiplied by 100 (percent per quarter)
Frequency / period	Quarterly, 1948:2 to 1983:4
Observations	143
Unit of observation	Aggregate national price index, one country
Filters, winsorization, standardization	None reported
Vintage	September 1984 publication; later benchmark revisions not reflected

Source: section 7 and footnotes 7–8 of the paper.

CRITIQUE Several data issues matter for the interpretation. The sample spans the Korean War price controls and their removal, the 1971–1974 wage and price controls and the Volcker disinflation, so the variance process is plausibly subject to regime change. The deflator is a revised series: the analysis models uncertainty about final inflation, not the real-time uncertainty faced by agents, which weakens the adaptive-learning interpretation. The sample is short for second-moment inference: 143 observations is small for estimating three variance parameters, a point taken up in Section 8. Survivorship and look-ahead bias in the usual cross-sectional sense do not arise.

4.2 Empirical Strategy and Identifying Variation

The empirical design is a sequence of nested specification checks rather than a causal design. First, a univariate AR(4) is identified with Box–Jenkins methods and estimated by OLS with constant variance. Second, the residuals are tested for ARCH through squared-residual autocorrelations and LM tests. Third, the Engle–Kraft ARCH(8) with fixed linear weights is re-estimated by ML. Fourth, LM tests probe both the linear restriction and the omission of h_{t-1} . Fifth, a GARCH(1,1) is estimated and subjected to the same diagnostics plus LM tests for higher-order GARCH and for the ARCH(8) lag.

INFERENCE The identifying variation for the variance parameters is the serial dependence in squared AR residuals: periods of large forecast errors that follow other large errors identify α_1 , while the slowness with which the effect decays identifies β_1 . With one series and 143 points, the variation is dominated by a few episodes: the volatile late 1940s and early 1950s, the calm 1960s and the post-1973 period. The design can establish that a GARCH(1,1) gives an adequate statistical description of these episodes. It cannot establish that the clustering is generated by a stationary learning process rather than by a small number of shifts in the variance level.

5 Results

5.1 Three Estimated Models

Table 3. Estimated models for quarterly U.S. GNP deflator inflation, 1948:2 to 1983:4

Parameter	(29) OLS, constant variance	(30) ARCH(8), linear lag	(31) GARCH(1,1)
Constant	0.240 (0.080)	0.138 (0.059)	0.141 (0.060)
π_{t-1}	0.552 (0.083)	0.423 (0.081)	0.433 (0.081)
π_{t-2}	0.177 (0.089)	0.222 (0.108)	0.229 (0.110)
π_{t-3}	0.232 (0.090)	0.377 (0.078)	0.349 (0.077)
π_{t-4}	-0.209 (0.080)	-0.175 (0.104)	-0.162 (0.104)
Variance intercept	$h = 0.282 (0.034)$	0.058 (0.033)	0.007 (0.006)
ARCH weight	n.a.	0.802 (0.265) on $\sum(9-i)/36 \cdot \varepsilon_{t-i}^2$	$\alpha_1 = 0.135$ (0.070)
GARCH weight	n.a.	n.a.	$\beta_1 = 0.829$ (0.068)
Kurtosis of standardized residuals	6.90	4.07	3.81

Source: equations (29), (30), (31) and section 7 of the paper. OLS standard errors in column (29); heteroskedasticity-consistent ML standard errors in columns (30) and (31). Asymptotic standard error of sample kurtosis under normality, $2\sqrt{(24/T)}$ for a two-standard-error band, equals 0.82.

RESULT 1: CONSTANT-VARIANCE AR(4) LEAVES STRONG ARCH STRUCTURE

EVIDENCE The OLS AR(4) is stationary and its residuals show no significant autocorrelation at the first ten lags. Squared residuals, however, have significant autocorrelations at lags 1, 3, 7, 9 and 10, similar partial autocorrelations, and LM tests for ARCH(1), ARCH(4) and ARCH(8) are highly significant. Standardized-residual kurtosis is 6.90. The mean is well specified, the variance is not: interval forecasts from (29) are wrong even though point forecasts are fine.

RESULT 2: ARCH(8) CAPTURES MOST OF THE CLUSTERING BUT REJECTS ITS OWN RESTRICTION

EVIDENCE The ARCH(8) total weight of 0.802 is statistically significant ($t \approx 3.0$). Fourth-moment existence under Milhøj's condition is just met, with a value of 0.989. Diagnostics on standardized residuals are clean. The LM test of the linear restriction gives 8.87, the 0.74 fractile of its χ^2 distribution, so the restriction is not rejected. The LM test for adding h_{t-1} gives 4.57, significant at 5 percent. The data thus accept the linear shape but want a longer tail than eight quarters. The implied unconditional variance, $0.058/(1 - 0.802) = 0.293$ (reviewer's computation), is close to the OLS residual variance of 0.282.

RESULT 3: GARCH(1,1) PASSES EVERY REPORTED DIAGNOSTIC

EVIDENCE In (31) the mean coefficients are almost identical to the ARCH model, consistent with block-diagonality of the information matrix. The variance equation gives $\alpha_1 = 0.135$ ($t \approx 1.9$) and $\beta_1 = 0.829$ ($t \approx 12.2$); persistence is 0.964 and the fourth moment exists by Theorem 2. No standardized-residual or squared standardized-residual autocorrelation exceeds two standard errors. The LM test for adding the ARCH(8) lag is 2.33 (0.87 fractile), for GARCH(1,2) or the locally equivalent GARCH(2,1) it is 3.80 (not significant at 5 percent), and for adding ε_{t-2}^2 to ε_{t-5}^2 it is 5.58 (0.77 fractile). Standardized-residual kurtosis falls to 3.81, within roughly two asymptotic standard errors of normality, and the three skewness values reported (−0.13, 0.18 and 0.11) are all within one standard error of 0.20.

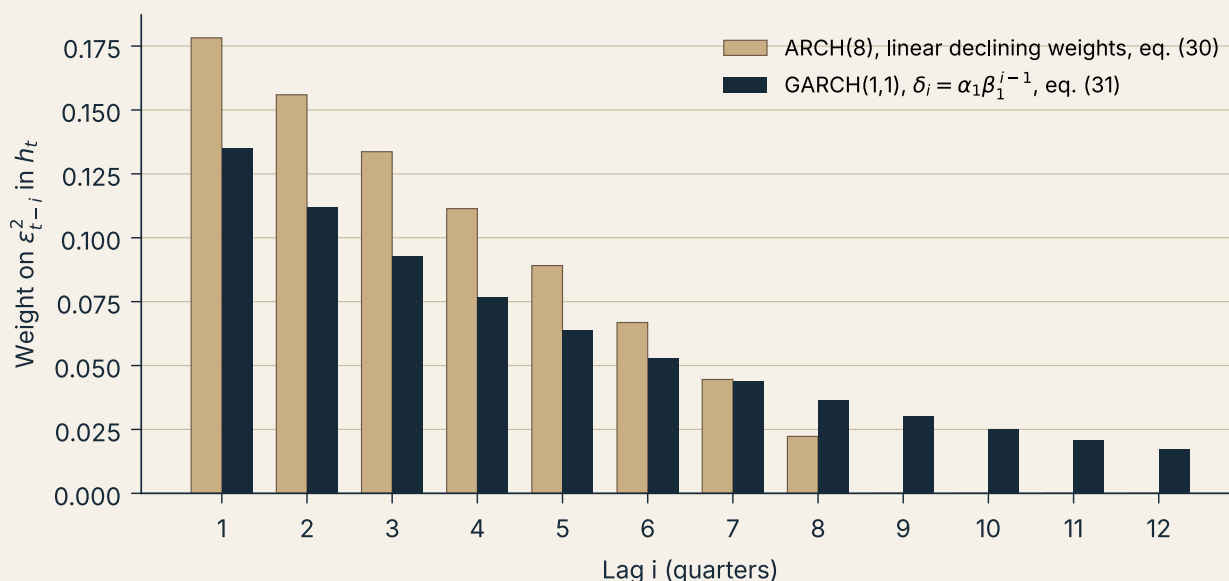
CRITIQUE The “marginally better fit” is asserted, not measured: the paper reports no log-likelihood values and no information criteria, and the two models are non-nested. The comparison rests on the asymmetric LM evidence: h_{t-1} is significant when added to ARCH(8), while the ARCH(8) term is not when added to GARCH(1,1). That asymmetry is the strongest piece of evidence in the paper, although with α_1 significant only at about the 6 percent level, the precise split between α_1 and β_1 is not tightly pinned down.

RESULT 4: A LONGER AND MORE PLAUSIBLE MEMORY

EVIDENCE The GARCH(1,1) mean lag is 5.848 quarters and its median lag 3.696 quarters, against a mean lag forced to $3\frac{1}{3}$ and a median lag of $2\frac{1}{2}$ in the ARCH(8). Both GARCH figures reproduce exactly from (L): $1/(1 - 0.829) = 5.848$ and $-\ln 2/\ln 0.829 = 3.696$.

INFERENCE Several economically meaningful quantities follow from (31) but are not reported. The half-life of a variance shock, $\ln 0.5/\ln(\alpha_1 + \beta_1)$, is about 18.9 quarters, so roughly half of an uncertainty shock is still present after almost five years. The implied unconditional variance is $0.007/(1 - 0.964) \approx 0.194$, an unconditional standard deviation of about 0.44 percentage points per quarter. The implied unconditional kurtosis from (K2) is about 6.19, close to the 6.90 sample kurtosis of the constant-variance residuals. GARCH(1,1) therefore reproduces the fat tails of raw inflation surprises almost entirely through volatility clustering, a validation the paper could have reported but did not.

Figure 2. Weights on lagged squared innovations in the conditional variance: ARCH(8) of equation (30) versus GARCH(1,1) of equation (31)



Generated by the reviewer from the published point estimates; this redraws the content of the paper's Fig. 2. Units: weight per unit of squared innovation (percent per quarter, squared). GARCH weights continue beyond lag 12, summing to $\alpha_1/(1 - \beta_1) = 0.789$.

Figure 2 shows why the two models fit almost equally well and yet differ economically. Their total weights are nearly identical, 0.802 for ARCH(8) and 0.789 for GARCH(1,1). Over lags 1 to 4 the ARCH(8) puts more weight on recent news; beyond lag 6 the GARCH tail dominates, and beyond lag 8 the ARCH(8) is forced to zero. The LM test for h_{t-1} in the ARCH model is detecting exactly this missing tail. The figure also shows the limit of GARCH(1,1): its weights are monotone, so it cannot represent a hump-shaped response of uncertainty to news.

5.2 Figures 3 and 4 of the Paper: Interval Forecasts

PAPER The paper's Figures 3 and 4 plot actual inflation with 95 percent asymptotic intervals for one-step-ahead forecast errors from the OLS model (29) and the GARCH model (31). The GARCH intervals are wide from the late 1940s to the mid-1950s, when inflation was volatile and hard to predict; narrow in the 1960s and early 1970s, when the constant OLS interval appears much too wide; and somewhat wider after the 1974 oil shock, though never as wide as at the start of the sample.

CRITIQUE The figures are a visual demonstration, not a test. The paper reports no coverage rates, so it does not show whether the GARCH intervals contain about 95 percent of outcomes in each sub-period while the OLS intervals over-cover in the 1960s and under-cover early on. A conditional coverage test in the spirit of Christoffersen (1998) is the formal version of this argument. The figures also hint at the main threat to the interpretation: the variance appears to fall in steps between regimes rather than to fluctuate around a single stationary level.

6 Robustness and Additional Tests

The paper's robustness evidence consists of diagnostic and LM tests within the example, plus the quasi-likelihood remark in section 5. Each is evaluated below in the form concern, test, result and assessment.

Table 4. Robustness evidence reported in the paper and its adequacy

Concern	Test in the paper	Result	Does it resolve the concern?
Remaining serial correlation in the mean	First ten ACF and PACF of $\varepsilon_t h_t^{-1/2}$	None exceed two standard errors	Largely yes for linear dependence; portmanteau statistics would be preferable to ten individual tests
Remaining ARCH effects	ACF and PACF of $\varepsilon_t^2 h_t^{-1}$; LM for ε_{t-2}^2 to ε_{t-5}^2	Not significant; LM 5.58, 0.77 fractile	Yes within the class of linear variance equations; asymmetric (leverage) effects are not tested
Under-parameterised GARCH order	LM for GARCH(1,2) or GARCH(2,1)	3.80, not significant at 5 percent	Partially: by the paper's own result these two alternatives cannot be distinguished
ARCH(8) dominates GARCH	LM for adding the ARCH(8) lag to GARCH(1,1)	2.33, 0.87 fractile	Supportive, but a nesting test, not a non-nested comparison of fit
Non-normal conditional distribution	Kurtosis and skewness of standardized residuals	Kurtosis 3.81, skewness within one standard error	Mostly, with low power at $T = 143$; theory in (25) covers QMLE
Inference under non-normality	Heteroskedasticity-consistent standard errors (sandwich form)	Reported for (30) and (31)	Yes in principle, although QMLE theory for GARCH was not yet established
Parameter stability and structural breaks	None	Not tested	No; this is the central omitted check (Section 7)
Out-of-sample forecasting	None	Not tested	No; the comparison is entirely in-sample

Source: section 7 of the paper; assessments by the reviewer.

CRITIQUE The robustness evidence is appropriate for a methodological paper whose example is illustrative, but it does not address the two threats most relevant to the economic claim: instability of the variance process across policy regimes, and out-of-sample forecast performance. The diagnostics also share a common weakness: each is a separate significance test at the 5 percent level, with no correction for the number of tests and low power in a sample of 143.

7 Critical Evaluation

7.1 What the Paper Actually Proves

ESTABLISHED

The GARCH(p,q) process is well defined under non-negativity constraints; it is covariance stationary if and only if $A(1) + B(1) < 1$, with unconditional variance $\alpha_0/(1 - A(1) - B(1))$; it has an exact ARCH(∞) representation and an exact ARMA(m, p) representation in ε_t^2 ; the GARCH(1,1) has the moment conditions (9) and the kurtosis (K2), and is leptokurtic whenever $\alpha_1 > 0$; under a white-noise null the information matrix for a full GARCH(p,q) alternative is singular.

SUPPORTED BUT CONDITIONAL

ML estimation is consistent and asymptotically normal with a block-diagonal information matrix, and the sandwich covariance applies under (25). These results hold by analogy with Weiss (1982) and were proved rigorously for GARCH only later. The ACF and PACF identification tools require finite fourth moments.

SUGGESTIVE

GARCH(1,1) describes U.S. inflation uncertainty slightly better than the restricted ARCH(8), and its lag structure is more plausible. The evidence is in-sample, uses one series and 143 observations, and reports no likelihood comparison.

NOT ESTABLISHED

That inflation uncertainty follows an adaptive learning mechanism; that the estimated persistence reflects genuine long memory rather than regime shifts; that GARCH improves out-of-sample interval forecasts; that conditional normality is the right distributional assumption. The learning interpretation is a reading of the functional form, not something the data test.

The distinctions that matter here are model result versus empirical fact and statistical description versus mechanism. Everything in sections 2 to 6 of the paper is a mathematical result and holds unconditionally. Everything in section 7 is a statistical description of one series, and the move from “the lagged variance term is significant” to “uncertainty is formed by adaptive learning” is an interpretation the empirical design cannot support.

7.2 Strengths

STRENGTHS RECOGNISED BY THE AUTHOR

- Parsimony through the ARMA analogy: a long, non-negative lag from three parameters removes the need for arbitrary lag restrictions, which was the explicit practical problem.

- Tractability: estimation and testing reuse the ARCH machinery with one added recursion, and the information matrix stays block diagonal, so existing regression software extends naturally.
- Diagnostic tools: ACF and PACF of squared residuals give an identification method familiar from Box and Jenkins.

STRENGTHS IDENTIFIED INDEPENDENTLY

- Exact, checkable theory. Theorems 1 and 2 are necessary and sufficient, not merely sufficient, and they give closed-form unconditional moments. This makes GARCH one of the few nonlinear time-series models whose moment structure can be computed by hand, which is why it became the default.
- Separation of scale and shape. In GARCH(1,1), α_1 controls reactivity and tail thickness, while β_1 controls the shape of memory through (L). Two parameters map onto two interpretable features of the data, which explains the model's empirical robustness across asset classes.
- Honest treatment of testability. The infeasibility result was inconvenient for the paper's own model and could easily have been omitted; reporting it anticipated an important literature on non-standard testing.
- Foresight on quasi-likelihood. The remark around (25) anticipates the QMLE practice that made GARCH robust to fat-tailed conditional distributions.

7.3 Weaknesses

REGIME SHIFTS CAN MASQUERADE AS PERSISTENCE

CRITIQUE Problem: a variance process with a few discrete level shifts, such as high in 1948–1955, low in the 1960s and higher after 1973, generates long-lasting autocorrelation in squared residuals that a stationary GARCH can only fit by setting $\alpha_1 + \beta_1$ close to one. **Why it matters:** estimated persistence, half-life and unconditional variance would then be artefacts. **Potential bias:** upward bias in β_1 and persistence, as documented by Diebold (1986) and Lamoureux and Lastrapes (1990). The paper's own numbers hint at this: the GARCH-implied unconditional variance of 0.194 is about 31 percent below the sample residual variance of 0.282, consistent with an early high-variance regime that the stationary model treats as transitory. **How to test:** Chow-type or Andrews sup-LM tests on the variance parameters, estimating on sub-samples split at 1955 and 1973, or Markov-switching GARCH (Gray, 1996).

SMALL SAMPLE FOR SECOND-MOMENT INFERENCE

CRITIQUE Problem: 143 quarterly observations carry little information about a near-integrated variance process. **Why it matters:** standard errors from asymptotic theory may badly understate uncertainty about α_1 , β_1 and their sum. Section 8 quantifies this through simulation. **How to test:** parametric bootstrap of the likelihood ratio or of persistence, or Bayesian estimation with posterior summaries.

NO FORMAL MODEL COMPARISON

CRITIQUE Problem: the superiority claim rests on LM tests and visual plausibility, with no likelihoods, information criteria or forecast evaluation. Why it matters: two models with nearly equal total lag weight (0.789 versus 0.802) may be indistinguishable in fit. How to test: report log-likelihoods and AIC/BIC; use a Vuong (1989) non-nested test; compare out-of-sample variance forecasts with robust loss functions (Patton, 2011).

SYMMETRIC NEWS RESPONSE AND POSITIVITY CONSTRAINTS

CRITIQUE Problem: h_t responds to ε_{t-1}^2 , so a positive and a negative surprise of equal size have identical effects, and non-negativity requires sign constraints. Why it matters: for equities the leverage effect (Black, 1976) makes negative shocks raise volatility more, and for inflation upside and downside surprises may have different uncertainty implications. How to test: GJR-GARCH (Glosten, Jagannathan and Runkle, 1993) or EGARCH (Nelson, 1991), with sign-bias tests (Engle and Ng, 1993).

THE LEARNING INTERPRETATION IS UNTESTED

CRITIQUE Problem: the functional form resembles adaptive expectations, but the paper offers no economic model in which agents form variance forecasts this way, and uses revised rather than real-time data. Why it matters: the economic claim that uncertainty is learned adaptively is not identified from a reduced-form fit. How to test: compare GARCH variance forecasts with survey measures of inflation uncertainty, such as the dispersion or individual histograms of the Survey of Professional Forecasters, using real-time data vintages.

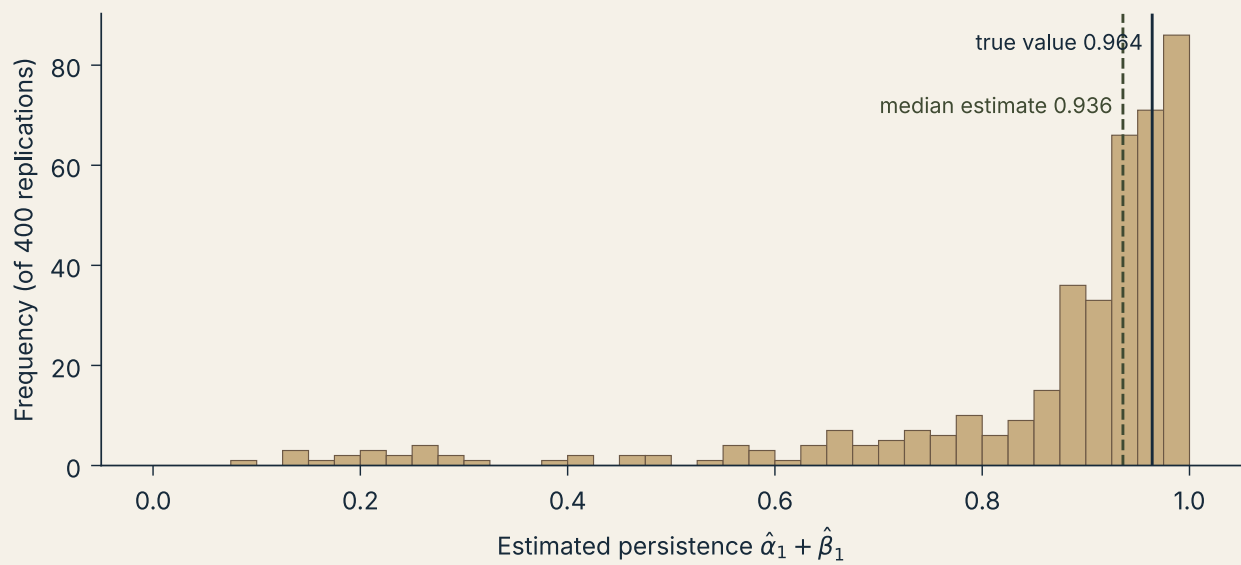
SINGLE MOST IMPORTANT WEAKNESS

The empirical example cannot separate genuine persistent volatility clustering from a small number of shifts in the variance level across policy regimes. Because the economic interpretation of GARCH, including persistence, half-life and learning, depends on this distinction, the inflation application demonstrates fit but not mechanism.

8 Contradictions and Edge Cases

8.1 How Precisely Can 143 Observations Pin Down Persistence?

EXTENSION To test the reliability of the paper's estimates, the reviewer simulated 400 samples of length 143 from a GARCH(1,1) with the paper's parameters ($\alpha_0 = 0.007$, $\alpha_1 = 0.135$, $\beta_1 = 0.829$, normal innovations, 500 burn-in observations) and re-estimated each by Gaussian ML. This is an illustrative calculation by the reviewer, not a result of the paper.

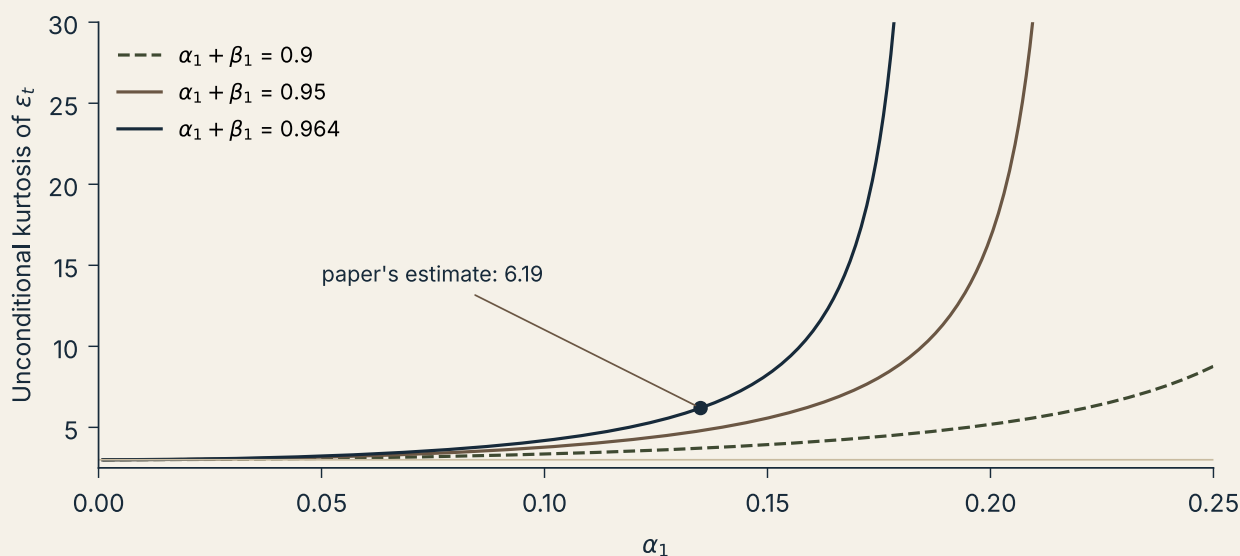
Figure 3. Sampling distribution of estimated persistence $\hat{\alpha}_1 + \hat{\beta}_1$ at $T = 143$ under the paper's GARCH(1,1) estimates

Generated by the reviewer: Monte Carlo with 400 replications, Gaussian ML (Nelder–Mead, three starting values, stationarity imposed), recursion initialised at the sample variance. Estimates at the stationarity boundary are shown at 1.

The median estimate of persistence is 0.936, and the 5th to 95th percentile range runs from about 0.40 to the stationarity boundary. 12.5 percent of replications return persistence above 0.99. $\hat{\alpha}_1$ ranges from about 0.02 to 0.30 and $\hat{\beta}_1$ from about 0.19 to 0.94 across the same percentiles, with 12 percent of replications estimating $\hat{\beta}_1$ below 0.5. Two conclusions follow. The reported standard errors of 0.070 and 0.068 understate the uncertainty about the decomposition of persistence. The sampling distribution is skewed and piles up against the unit boundary, so asymptotic normal intervals are unreliable at this sample size. The qualitative conclusion that inflation uncertainty is persistent survives; the specific mean lag of 5.85 quarters and half-life of 19 quarters do not deserve the precision their decimals suggest.

8.2 Kurtosis and the Fragility of Moment-Based Tools

Figure 4. Unconditional kurtosis of a Gaussian GARCH(1,1) as a function of α_1 at fixed persistence



Generated by the reviewer from equation (K2): $kurtosis = 3 + 6\alpha_1^2/(1 - \beta_1^2 - 2\alpha_1\beta_1 - 3\alpha_1^2)$, with $\beta_1 = \text{persistence} - \alpha_1$. Curves end where the fourth moment ceases to exist.

At persistence 0.964, kurtosis is about 6.2 at the paper's $\alpha_1 = 0.135$ but diverges as α_1 approaches about 0.15, where the fourth-moment condition fails. The paper's estimate therefore sits on a steep part of the surface: a change in α_1 well within one standard error moves the model from finite to infinite kurtosis. This has two edge-case implications. The ACF and PACF tools of section 4 of the paper, which require finite fourth moments, are on fragile ground for this very example. Sample kurtosis converges extremely slowly near the boundary: in the simulation above, the median sample kurtosis of raw ε_t at $T = 143$ was only 3.2, against a population value of 6.2. The 6.90 sample kurtosis of the OLS residuals is thus, if anything, higher than a typical draw from the fitted GARCH would produce, consistent with the presence of an additional source of tail mass such as regime shifts.

8.3 When the Mechanism Fails or Changes

- Persistence equal to one. When $\alpha_1 + \beta_1 = 1$, Theorem 1 fails, the unconditional variance is infinite and shocks to the variance forecast never decay (IGARCH; Engle and Bollerslev, 1986). Yet the process remains strictly stationary under Nelson's (1990) condition, so the boundary is a statement about second moments, not about explosive behaviour. The paper's framework provides no guidance here.
- $\alpha_1 = 0$. Then β_1 is unidentified and the LM infeasibility of section 6 bites. In finite samples, near this point, ML estimates of β_1 become erratic, which is visible in the left tail of Figure 3.
- Negative coefficients in higher-order models. The constraints $\alpha_i, \beta_i \geq 0$ are sufficient but not necessary for $h_t > 0$ when p or q exceeds one; Nelson and Cao (1992) give weaker necessary and sufficient conditions. Imposing the paper's constraints can therefore exclude valid models.

- Aggregation. GARCH is not closed under temporal aggregation: a GARCH(1,1) at daily frequency does not imply a GARCH(1,1) at quarterly frequency (Drost and Nijman, 1993). A quarterly GARCH for inflation is a reduced-form approximation whose parameters have no invariant meaning across frequencies.
- Alternative mechanisms generating the same fit. A stochastic volatility model, a Markov-switching variance, or a variance driven by an observed policy-regime indicator can all produce the same squared-residual autocorrelations. The LM diagnostics in the paper cannot distinguish these.

9 Further Literature

Table 5. Literature connections: paper, shared idea, relationship and why it matters

Paper	Shared idea	Relationship	Why it matters
Engle (1982)	Conditional variance as a function of past squared shocks	Direct foundation; GARCH nests ARCH(q)	Defines the problem GARCH solves and the inference framework it extends
Box and Jenkins (1976)	ARMA parsimony; identification via ACF and PACF	Methodological template transplanted to the second moment	Explains why GARCH is to ARCH what ARMA is to AR
Mandelbrot (1963)	Fat tails and clustering of large price changes	Stylised facts that GARCH explains via (K2)	Links GARCH to the earlier stable-Paretian interpretation of fat tails
Engle and Bollerslev (1986)	Persistence of variance	Extension to the unit-persistence case, IGARCH	Addresses near-unit persistence typical of financial data
Bollerslev (1987)	Conditional distribution	Student-t GARCH for speculative prices and rates of return	Responds to residual excess kurtosis left by normal GARCH
Engle, Lilien and Robins (1987)	Variance as a risk premium	ARCH-in-mean puts h_t in the mean equation	Connects GARCH to asset pricing and the risk-return trade-off
Bollerslev, Engle and Wooldridge (1988)	Time-varying covariances	Multivariate GARCH in a CAPM setting	Opens multivariate volatility and conditional beta modelling
Nelson (1990)	Stationarity of GARCH(1,1)	Strict stationarity and ergodicity beyond Theorem 1	Corrects the reading of persistence near one
Nelson (1991)	Asymmetric news response	EGARCH; log-variance avoids sign constraints	Addresses symmetry and positivity limitations

Paper	Shared idea	Relationship	Why it matters
Glosten, Jagannathan and Runkle (1993)	Leverage effect	Threshold GARCH with asymmetric α	Most widely used asymmetric extension in equity markets
Diebold (1986); Lamoureux and Lastrapes (1990)	Persistence estimates	Show structural shifts inflate measured persistence	Directly challenges the interpretation of the inflation example
Bollerslev and Wooldridge (1992)	Quasi-maximum likelihood	Robust standard errors under non-normality	Formalises the remark around (25); now standard practice
Lee and Hansen (1994); Lumsdaine (1996)	Asymptotics of the QMLE	Rigorous consistency and normality for GARCH(1,1), including IGARCH	Supplies the proofs that the paper borrowed by analogy
Francq and Zakoïan (2004)	Asymptotics of the QMLE	General GARCH(p,q) theory under mild conditions	Completes the estimation theory of the model
Baillie, Bollerslev and Mikkelsen (1996)	Long memory in volatility	Fractionally integrated GARCH with hyperbolic decay	Responds to the geometric memory limit of GARCH(1,1)
Andersen and Bollerslev (1998)	Forecast evaluation	Shows GARCH forecasts are good when judged against realised volatility	Reverses the apparent low R^2 of volatility forecasts
Hansen and Lunde (2005)	Benchmark comparison	330 ARCH-type models fail to beat GARCH(1,1) for exchange rates	Empirical confirmation of the paper's parsimony argument
Engle (2002)	Scalable multivariate volatility	DCC: univariate GARCH plus correlation dynamics	Industrial use of GARCH in risk management
Hansen, Huang and Shek (2012)	Measurement of the variance	Realized GARCH adds high-frequency measures	Modern update of the information set in h_t

Source: reviewer's synthesis; full references in the bibliography.

Two connections deserve emphasis. Hansen and Lunde (2005) is the strongest later evidence for the paper's parsimony argument: across hundreds of more flexible specifications, GARCH(1,1) is rarely beaten out of sample for exchange rates, though leverage models do better for equities. Diebold (1986) and Lamoureux and Lastrapes (1990) are the strongest challenge: they show that the persistence GARCH measures is partly a symptom of structural instability, which is precisely the threat to the paper's own inflation example.

10 Research Gaps

10.1 Paper-Level Gaps

Gap: no likelihood comparison or forecast evaluation. Why it matters: the empirical superiority claim is unquantified. Existing limitation: in-sample LM tests only. Way forward: report likelihoods, information criteria and out-of-sample coverage. Gap: no general fourth-moment condition for GARCH(p,q); footnote 3 states that it is unknown. Why it matters: the identification tools require it. Way forward: closed-form conditions were later derived by He and Teräsvirta (1999) and Ling and McAleer (2002).

10.2 Literature-Level Gaps

Gap: the economic origin of volatility clustering. Why it matters: GARCH describes clustering without explaining it, so persistence parameters lack structural interpretation. Existing limitation: reduced-form models dominate. Way forward: link variance dynamics to information-arrival models (Clark, 1973; Andersen, 1996), learning models with parameter uncertainty, or policy-regime variables. Gap: separating persistence from breaks. Why it matters: risk forecasts and option prices depend on it. Way forward: change-point GARCH and time-varying unconditional variance models such as the spline-GARCH of Engle and Rangel (2008).

10.3 Methodological Gaps

Gap: finite-sample inference in short macro samples. Why it matters: Section 8 shows asymptotic intervals are unreliable at $T = 143$. Way forward: bootstrap or Bayesian inference, and boundary-aware tests (Andrews, 2001). Gap: testing GARCH directly against white noise. Why it matters: the paper proves standard tests are infeasible. Way forward: Davies-type sup tests or tests that exploit the one-sided nature of the alternative.

11 Research Opportunities

Idea 1. Regime-aware persistence of U.S. inflation uncertainty, 1947 to present

RESEARCH QUESTION. Does the high persistence of inflation uncertainty estimated by GARCH survive once the variance is allowed to shift across monetary-policy regimes?

MOTIVATION. Section 8 shows the paper's persistence is imprecise and possibly inflated by breaks; the post-2021 inflation episode provides a new out-of-sample regime.

HYPOTHESIS. Within regimes, $\alpha_1 + \beta_1$ is materially below 0.96, and most low-frequency variation is in the regime-specific unconditional variance.

ECONOMIC MECHANISM. Policy credibility anchors the long-run variance; within a credible regime, shocks to uncertainty decay quickly.

LITERATURE CONNECTION. Bollerslev (1986); Diebold (1986); Lamoureux and Lastrapes (1990); Gray (1996); Engle and Rangel (2008).

DATA. Quarterly GDP deflator and CPI from FRED and BEA, 1947 to latest; real-time vintages from the Philadelphia Fed Real-Time Data Set.

VARIABLES. Inflation, AR residuals, regime indicators (Fed chair, inflation-targeting announcement of 2012), oil-price shocks.

METHOD. GARCH(1,1), Markov-switching GARCH, spline-GARCH, and GARCH with Bai–Perron break dates in α_0 ; bootstrap inference.

IDENTIFICATION STRATEGY. Compare persistence estimates with and without regime structure; use pre-dated institutional break dates to avoid data-mined breaks.

EXPECTED CONTRIBUTION. A credible decomposition of inflation-uncertainty dynamics into regime level and transitory clustering.

MAIN RISK. Few regimes and few observations per regime; inference remains imprecise.

Idea 2. Does GARCH uncertainty match what forecasters say they are uncertain about?

RESEARCH QUESTION. Do real-time GARCH conditional variances of inflation track the subjective uncertainty in density forecasts from the Survey of Professional Forecasters?

MOTIVATION. The paper interprets GARCH as adaptive learning but never confronts it with agents' expectations.

HYPOTHESIS. Survey uncertainty loads on lagged survey uncertainty (β -like) and lagged squared surprises (α -like), with weights similar to real-time GARCH.

ECONOMIC MECHANISM. Adaptive updating of perceived variance from forecast errors.

LITERATURE CONNECTION. Bollerslev (1986); Engle (1983); survey-based uncertainty literature using SPF probability histograms.

DATA. SPF individual probability distributions for GDP-deflator inflation since 1968; real-time inflation vintages.

VARIABLES. Individual and consensus histogram variance, real-time GARCH h_t , realised squared forecast errors.

METHOD. Panel regressions of survey variance on GARCH-type regressors; Mincer-Zarnowitz regressions of squared errors on both measures.

IDENTIFICATION STRATEGY. Real-time data ensure GARCH uses only information available to forecasters; individual fixed effects absorb respondent-specific bias.

EXPECTED CONTRIBUTION. A direct test of the learning interpretation of GARCH.

MAIN RISK. Changes in SPF histogram bins and horizons complicate measurement.

Idea 3. Finite-sample reliability of GARCH in short macroeconomic samples

RESEARCH QUESTION. How large must T be before Gaussian QML intervals for persistence and half-life have correct coverage, and which alternative inference methods perform best below that threshold?

MOTIVATION. Figure 3 shows severe skewness and boundary pile-up at $T = 143$, yet quarterly macro GARCH remains common.

HYPOTHESIS. Asymptotic intervals under-cover substantially below a few hundred observations; parametric bootstrap and Bayesian intervals restore coverage.

ECONOMIC MECHANISM. Not economic; a statistical question about weak identification near $\alpha_1 = 0$ and persistence near one.

LITERATURE CONNECTION. Bollerslev (1986); Lumsdaine (1996); Andrews (2001); Francq and Zakoïan (2004).

DATA. Simulations calibrated to published macro GARCH estimates; empirical illustrations on inflation and GDP growth.

VARIABLES. Coverage rates, interval length, bias of persistence and half-life.

METHOD. Monte Carlo comparison of Wald, profile-likelihood, bootstrap and Bayesian (MCMC) intervals.

IDENTIFICATION STRATEGY. Known data-generating processes in simulation.

EXPECTED CONTRIBUTION. Practical guidance for applied macro and finance researchers using low-frequency data.

MAIN RISK. Results may be viewed as incremental unless tied to re-evaluating published findings.

Idea 4. When does GARCH(1,1) parsimony win for volatility-targeted portfolios?

RESEARCH QUESTION. Across asset classes and horizons, do GARCH(1,1) forecasts deliver better volatility-managed portfolio performance than asymmetric, long-memory or realised-measure models, after transaction costs?

MOTIVATION. Statistical forecast rankings need not translate into economic value; the parsimony argument of the paper should be judged by portfolio outcomes.

HYPOTHESIS. GARCH(1,1) is hard to beat for currencies and commodities, while asymmetric and realised-measure models add value for equity indices.

ECONOMIC MECHANISM. Leverage effects are strong in equities and weak in currencies; over-parameterised models add estimation noise that turnover costs amplify.

LITERATURE CONNECTION. Hansen and Lunde (2005); Glosten, Jagannathan and Runkle (1993); Hansen, Huang and Shek (2012); volatility-managed portfolio literature.

DATA. Daily futures returns for equity indices, bonds, currencies and commodities; high-frequency data where available.

VARIABLES. Forecast variance, realised variance, scaled portfolio returns, turnover, Sharpe ratio and drawdown.

METHOD. Rolling out-of-sample estimation; Patton (2011) robust losses; Diebold–Mariano and model confidence set tests; economic-value comparisons.

IDENTIFICATION STRATEGY. Strictly out-of-sample design with fixed estimation windows to prevent look-ahead.

EXPECTED CONTRIBUTION. A map of where GARCH(1,1) is sufficient and where it is not, in economic units.

MAIN RISK. Differences in Sharpe ratios may be statistically indistinguishable.

Idea 5. Inflation uncertainty in mean: revisiting the Friedman hypothesis with GARCH-M and modern identification

RESEARCH QUESTION. Does higher conditional inflation uncertainty depress subsequent real output growth once regime shifts and common shocks are controlled for?

MOTIVATION. The paper's application was motivated by inflation uncertainty as a variable of major importance but did not study its consequences.

HYPOTHESIS. Uncertainty shocks reduce output growth, and the effect is smaller under credible inflation targeting.

ECONOMIC MECHANISM. Uncertainty distorts relative-price signals and raises the option value of waiting for investment.

LITERATURE CONNECTION. Bollerslev (1986); Engle, Lilien and Robins (1987); Coulson and Robins (1985).

DATA. Quarterly inflation and real GDP for the United States and an inflation-targeting panel (OECD).

VARIABLES. Bivariate GARCH-M conditional variances, output growth, policy regime dummies.

METHOD. Bivariate VAR-GARCH-in-mean; cross-country panel with inflation-targeting adoption as a difference-in-differences shifter.

IDENTIFICATION STRATEGY. Timing restrictions within quarters plus staggered adoption of inflation targeting.

EXPECTED CONTRIBUTION. Evidence on whether the welfare channel of inflation uncertainty is regime dependent.

MAIN RISK. Endogeneity of inflation-targeting adoption and weak GARCH-M identification.

11.1 Ranking of Ideas

Table 6. Ranking of research ideas (scores from 1, weak, to 5, strong)

Idea	Novelty	Feasibility	Data	Method	Importance	Contribution	Total
2. GARCH versus survey uncertainty	4	4	4	4	4	4	24
1. Regime-aware inflation persistence	3	5	5	4	4	3	24
4. GARCH parsimony in volatility targeting	3	4	4	4	4	3	22
3. Finite-sample reliability	3	5	5	4	2	3	22
5. Uncertainty in mean	2	3	4	3	4	3	19

Source: reviewer's assessment.

The top three are Ideas 2, 1 and 4. Idea 2 is strongest because it directly tests the economic interpretation the paper offers but does not test, uses high-quality public data, and has a clean real-time design. Idea 1 is the most feasible and directly addresses the paper's single most

important weakness, with the post-2021 inflation surge as a natural new test. Idea 4 is broader than the paper but converts its parsimony argument into an economic question with immediate relevance for trading and risk management. Idea 4 ranks above Idea 3 on economic importance despite an equal total.

12 Learning Framework

12.1 Prerequisite Concepts

ARMA models and lag-polynomial algebra, including invertibility and roots outside the unit circle; conditional versus unconditional moments and the law of iterated expectations; martingale differences; moments of the normal distribution; maximum likelihood, the information matrix equality, the score and the outer-product estimator; Lagrange multiplier tests and their $T \cdot R^2$ form; and the Box–Jenkins identification cycle.

12.2 Core Concepts Learned

Volatility clustering as serial dependence in squared innovations; persistence $A(1) + B(1)$ as the governor of long-run variance; leptokurtosis generated by a conditionally Gaussian process; the $ARCH(\infty)$ and ARMA representations of GARCH; the separation between identification of shape (β_1) and scale (α_1); and the failure of standard tests when a parameter is identified only under the alternative.

12.3 Important Equations

The variance recursion (2); the $ARCH(\infty)$ form (4); the stationarity condition and unconditional variance of Theorem 1; the ARMA representation (6); the $GARCH(1,1)$ moment condition (9); the kurtosis formula (K2); the mean and median lags (L); and the Gaussian log-likelihood (18) with the recursive gradient (21).

12.4 Methods Learned

Recursive likelihood evaluation, BHHH optimisation, sandwich (QML) covariance estimation, LM testing through auxiliary regressions, and diagnostic checking with ACF and PACF of standardized and squared standardized residuals.

12.5 Questions to Test Understanding

COMPREHENSION

1. Why is $GARCH(1,1)$ equivalent to an $ARCH(\infty)$ with geometric weights, and what is the total weight?
2. What does Theorem 1 say about the unconditional variance as persistence approaches one?
3. Why is the $GARCH(1,1)$ process leptokurtic even though $\varepsilon_t | \psi_{t-1}$ is normal?

ANALYSIS

1. Derive the ARMA(1,1) representation of ε_t^2 for GARCH(1,1) and explain why OLS on it is inefficient.
2. Show that the mean lag depends only on β_1 , and explain what this implies for identification.
3. Explain why β_1 is unidentified when $\alpha_1 = 0$ and why this makes a general LM test for GARCH infeasible.

RESEARCH

1. Design a test that discriminates between a stationary GARCH and a GARCH with a single break in α_0 .
2. How would one test the adaptive-learning interpretation with survey data?
3. Under what data features would a long-memory volatility model dominate GARCH(1,1) out of sample?

12.6 Papers to Read Next

Engle (1982), for the original ARCH framework; Nelson (1990) and Nelson (1991), for stationarity and asymmetry; Bollerslev and Wooldridge (1992), for quasi-likelihood inference; Lamoureux and Lastrapes (1990), for the structural-change critique; Andersen and Bollerslev (1998), for forecast evaluation against realised volatility; and Hansen and Lunde (2005), for the benchmark status of GARCH(1,1).

13 Conclusion

ONE-SENTENCE THESIS

Adding lagged conditional variances to the ARCH equation yields a parsimonious, infinite-memory and analytically tractable model of time-varying volatility, the GARCH process, whose stationarity, moments, identification, estimation and testing can all be characterised with classical time-series tools.

THREE MOST IMPORTANT INSIGHTS

1. GARCH is to ARCH what ARMA is to AR: long memory in the variance from few parameters, with automatic non-negativity.
2. Persistence $A(1) + B(1)$ governs the long-run variance, while in GARCH(1,1) α_1 drives tail thickness and β_1 drives the shape of memory.
3. Fat-tailed unconditional distributions can arise entirely from time variation in a Gaussian conditional variance.

STRONGEST EVIDENCE

The exact theorems, especially Theorem 2 and the kurtosis formula, together with the asymmetric LM evidence in the inflation example: h_{t-1} is significant when added to ARCH(8), while the ARCH(8) lag is not when added to GARCH(1,1).

BIGGEST WEAKNESS

The empirical example cannot distinguish persistent volatility clustering from regime shifts in the variance level, and with 143 observations the persistence estimate is far less precise than the reported standard errors suggest.

MOST IMPORTANT RESEARCH GAP

An economic, rather than purely statistical, account of why conditional variances follow GARCH-like dynamics, and a method for separating that dynamic from structural change.

BEST RESEARCH OPPORTUNITY

Confronting real-time GARCH inflation uncertainty with the subjective uncertainty in Survey of Professional Forecasters density forecasts, within a regime-aware framework.

THREE PAPERS TO READ NEXT

Nelson (1991), Bollerslev and Wooldridge (1992), and Lamoureux and Lastrapes (1990).

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